Slipway: Accessing Finite Subspace Trails in Poseidon

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Abstract. Poseidon is an algebraic permutation designed for efficient use in proof systems. Its nonlinear layer consists of power-map S-boxes. In a full round, the S-box is applied to every state coordinate; in a partial round, it is applied to only one coordinate, reducing the arithmetization cost. Each round also applies an MDS linear layer to diffuse information across the state.

To study algebraic degree, we let the input depend on variables and follow the resulting family of states through the permutation. If the coordinate entering a partial-round S-box is constant across that family, the S-box adds no degree in the family variables. Directions with this property over several consecutive partial rounds form finite subspace trails. Such trails exist for every linear layer, but their existence does not by itself explain how a constrained family can pass through the preceding full rounds and enter them without first acquiring high degree.

We address this reachability problem by constructing a constrained input family and a round-constant-dependent MDS matrix together. The prescribed matrix images carry the family through the four initial full rounds and into a chosen finite trail. After an explicit change of variable, the state at the end of the full-round prefix is linear in the new root variable, so the prefix acts as a controlled reparametrization rather than as a source of degree growth. We call this effect *full-round absorption*.

For the KoalaBear instance $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$, we construct a two-parameter family whose first two input coordinates are zero. On this family, the four initial full rounds act as a reparametrization and deliver the variable directions into a two-dimensional trail, so those four rounds and the next fourteen partial S-boxes add no degree. For the exhibited control, the polynomials representing the first two output coordinates have exact degree $3^{R_F+R_P-4-14} = 3^{10}$, rather than the expected degree $3^{R_F+R_P} = 3^{28}$. We exhibit a common base-field root, yielding a complete CICO-2 solution for the full-round Poseidon instance.

The resulting matrices are MDS and satisfy the relevant matrix checks prescribed by the Poseidon designers, yet they make a finite trail reachable through the full-round prefix. We generalize the construction to CICO-k, derive bounds on trail dimension and matrix images, and provide a concrete MDS matrix that meets the CICO-3 matrix-image bound with equality.

Keywords: Poseidon · CICO · algebraic cryptanalysis · finite subspace trails · MDS matrices · zero-knowledge

1 Introduction

Proof systems over prime fields make native field operations comparatively inexpensive, while bit-oriented operations can require many constraints. This cost model has motivated a family of algebraic hash functions whose round functions are designed specifically for efficient arithmetization. Poseidon is a prominent example [7]. Following the HADES strategy [9], it surrounds a long sequence of inexpensive *partial rounds* by two shorter sequences of *full rounds*. A full round applies a nonlinear S-box to every state coordinate; a partial round applies it to only one coordinate. Each round then uses an MDS linear layer to diffuse information across the state.

To study algebraic degree, one replaces a single input state by a parameterized collection of states. Concretely, the input coordinates are written as expressions in variables $x_1, \ldots, x_d$, and applying the permutation carries this collection through a corresponding family at every intermediate round. We call the variables *family parameters* and measure degree in them. Fixing some parameters selects a lower-dimensional subfamily, or fiber; in particular, fixing all but one turns the final constraints into univariate polynomials.

If the coordinate entering the only active S-box has the same value for every state in the family, then the S-box acts on a constant and adds no degree in the family parameters. While this remains true, the variable-dependent directions are transported only by the linear layer. The directions that keep the active coordinate constant over several consecutive rounds form a finite subspace trail. For a width-t state, the space of directions that remain inactive for r partial rounds has dimension at least $t - r$; hence every linear layer admits a nonzero trail for up to $t - 1$ rounds. Beyne et al. exploited this finite-trail phenomenon in HadesMiMC, using a suitable input subspace to linearize consecutive partial rounds and derive integral properties [5]. Poseidon derives the corresponding dimension bound [7, Section 2.3, Equation (1)], and Li and Wang give an equivalent block-matrix characterization and describe when the bound is attained [12]. Finite trails are therefore unavoidable, and their mere existence is not evidence of a degenerate linear layer.

A trail only describes what happens after a family reaches the partial-round segment in a suitable direction. It does not explain how a constrained input family can pass through the preceding nonlinear full rounds and enter such a trail without first acquiring high degree. Previous analyses naturally fix the linear layer and characterize the trails it supports. We study the complementary reachability problem that arises when the linear layer may instead be chosen adaptively as a function of the round constants.

This adaptive freedom has a concrete motivation. Under the Ethereum Foundation's 2026 Poseidon Cryptanalysis Initiative bounty program, the round constants were fixed, whereas participants could select an arbitrary MDS matrix for the Poseidon-31 instance provided that it satisfied the program's "no-invariant-subspace-trail" conditions [6]. We retain the reference Grain LFSR round constants and use only the permitted freedom to select the matrix. Although our construction targets the full-round Poseidon instance rather than one of the

bounty's reduced-round challenges, this permitted freedom motivates an adaptive search for weak matrices. This raises the question: can a matrix pass the prescribed matrix-level checks, yet depend on the fixed constants in such a way that it steers a constrained input family through the initial full rounds and into a finite trail?

We answer affirmatively by constructing the input family and the matrix together. We prescribe the action of the matrix on independent vectors that represent the constant and parameter-dependent parts of the intermediate states. These prescribed images separate the parameter supports as the family crosses the four initial full rounds. After the explicit change of variable $X = x^{27}$, the state at the end of this prefix is linear in X, so the prefix acts as a controlled reparametrization rather than as four successive sources of degree growth. We call this effect *full-round absorption*. The resulting variable directions are placed in the two-dimensional space that keeps the active coordinate constant for the next fourteen partial rounds. Thus two distinct mechanisms remove degree growth: the change of variable absorbs the full-round prefix, while the finite trail neutralizes the subsequent partial S-boxes. In particular, the construction does not create an exceptional eighteen-round partial-round trail.

Our concrete target is to solve a constrained-input–constrained-output problem with two constrained coordinates (CICO-2): the first two coordinates of both the input and the output must be zero. For Poseidon over the KoalaBear field $2^{31} - 2^{24} + 1$, with parameters $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$, we construct a two-parameter family whose first two input coordinates vanish identically. Because the cubic map is bijective over this field, replacing x by $X = x^{27}$ does not lose any parameter values. Once the second parameter is fixed to an exhibited control value, the polynomials representing the first two output coordinates become univariate in X. They have exact degree $3^{R_F+R_P-4-14} = 3^{10} = 59049$, rather than the direct degree bound $3^{R_F+R_P} = 3^{28}$, and we exhibit an explicit common root. This gives a complete CICO-2 solution for the full-round Poseidon instance.

Using an analogous construction with $R_F = 6$, we also obtain a solution to the bounty's zero-test challenge. The challenge interprets the $t = 16$ input as the eight coefficients of a degree-seven polynomial $P \in \mathbb{F}_{p^2}[Z]$, takes the first pair of hash outputs as $a_0 \in \mathbb{F}_{p^2}$, and asks for $P(a_0) = 0$ [6].

The matrix used for the CICO-2 solution passes the matrix requirements of the Poseidon Cryptanalysis Initiative. It is MDS, meaning that every square submatrix is nonsingular, and it has maximal partial-round rank growth, so the dimensions of its finite trails meet the generic lower bound exactly. Moreover, the characteristic polynomial of M^i is irreducible over the base field for every $1 \leq i \leq 64$, which proves that none of these powers has a nonzero proper invariant subspace. These checks rule out the matrix-level degeneracies they target, but do not prevent the matrix from encoding relations with the already fixed round constants. The result thus identifies a class of round-constant-dependent weak matrices; it is not an attack on a fixed standardized Poseidon matrix. Section 6.1 explains why the irreducibility computation satisfies the sufficient branch of the matrix verifier required by the Initiative.

Finally, we generalize the mechanism to CICO-k and derive two dimension limits. To address k output constraints, the balanced construction uses the root variable together with $k-1$ controls, whose post-prefix directions span a k-dimensional space. Under maximal rank growth, the space of directions that remain inactive for r partial rounds has dimension $t-r$, and therefore $r \leq t-k$. Constructing the matrix imposes an independent limit: absorbing $R_f = R_F/2$ initial full rounds prescribes its action on $R_f(k+1)-1$ independent vectors, while one further matrix image is reserved to complete the partial-round trail. Since a linear map on $\mathbb{F}^t$ has only t independently assignable images, this requires $R_f(k+1) \leq t$. For $t = 16$ and $R_F = 8$ (that is, $R_f = 4$), the direct construction therefore supports at most CICO-3. We give a second MDS matrix for this case and verify the exact degree reduction of its three constrained output polynomials, although we did not search for a common root for that system.

Organization Section 2 defines Poseidon, CICO, and finite trails, and places the result relative to earlier algebraic attacks. Section 3 gives the CICO-2 construction and its complete solution. Section 4 presents the analogous CICO-3 construction. Section 5 proves the general trail-dimension and matrix-image bounds and recalls the prior-art solving step. Section 6 discusses matrix validation and countermeasures. Exact construction data and the expanded CICO-3 prefix calculation appear in the appendices.

2 Background and Related Work

2.1 Notation

Let $\mathbb{F} = \mathbb{F}_p$ be a prime field. Poseidon permutation states are represented as column vectors in $\mathbb{F}^t$, and coordinates are indexed $0, \ldots, t-1$. The vector e_i is the i-th standard basis vector, so $e_i^T \boldsymbol{v}$ is coordinate i of $\boldsymbol{v}$. For a vector $\boldsymbol{v}$, the notation $\boldsymbol{v}^{\circ\alpha}$ means componentwise exponentiation:

$$\boldsymbol{v}^{\circ\alpha} = (v_0^\alpha, \ldots, v_{t-1}^\alpha)^T.$$

We use bold lower-case letters for vectors and calligraphic upper-case letters for named vector spaces. Matrices are written with ordinary upper-case letters. All spans, kernels, ranks, and dimensions are over $\mathbb{F}$ unless an extension field is explicitly specified.

The support and Hamming weight of $\boldsymbol{v}$ are

$$\operatorname{supp}(\boldsymbol{v}) = \{i : v_i \neq 0\}, \qquad \operatorname{wt}(\boldsymbol{v}) = |\operatorname{supp}(\boldsymbol{v})|.$$

We write $\operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ for linear span and $\mathrm{GL}_t(\mathbb{F})$ for the invertible $t \times t$ matrices.

A matrix $M \in \mathbb{F}^{t\times t}$ is *MDS* if every nonempty square submatrix of M is nonsingular.

The MDS condition is equivalent [13] to the branch-number condition

$$\operatorname{wt}(\boldsymbol{v}) + \operatorname{wt}(M\boldsymbol{v}) \geq t+1 \quad \text{for every } \boldsymbol{v} \neq 0.$$

2.2 The Poseidon Permutation

We give the round definition explicitly because the location of each constant and matrix multiplication matters to the construction. Fix an S-box exponent α with $\gcd(\alpha, p-1) = 1$, a matrix $M \in \mathrm{GL}_t(\mathbb{F})$, and round constants $\boldsymbol{c}^{(r)} \in \mathbb{F}^t$. Define

$$S_F(\boldsymbol{z}) = \boldsymbol{z}^{\circ\alpha}, \qquad S_P(\boldsymbol{z}) = (z_0^\alpha, z_1, \ldots, z_{t-1})^T.$$

A full round and a partial round are, respectively,

$$\begin{aligned} R_r^F(\boldsymbol{s}) &= MS_F(\boldsymbol{s} + \boldsymbol{c}^{(r)}), \\ R_r^P(\boldsymbol{s}) &= MS_P(\boldsymbol{s} + \boldsymbol{c}^{(r)}). \end{aligned} \tag{1}$$

Poseidon applies $R_f = R_F/2$ full rounds, then R_P partial rounds, then another R_f full rounds:

$$\pi = R^F_{R_F+R_P-1} \circ \cdots \circ R^F_{R_f+R_P} \circ R^P_{R_f+R_P-1} \circ \cdots \circ R^P_{R_f} \circ R^F_{R_f-1} \circ \cdots \circ R^F_0.$$

For an input state $\boldsymbol{s}$, define the *initial full-round prefix* and its output by

$$\mathrm{Prefix}_{R_f} := R^F_{R_f-1} \circ \cdots \circ R^F_0, \qquad \boldsymbol{q}_{R_f} := \mathrm{Prefix}_{R_f}(\boldsymbol{s}).$$

Thus the prefix consists exactly of the first $R_f = R_F/2$ full rounds, including their constant additions, S-box layers, and matrix multiplications as defined in (1). The relevant order of states and round segments is

$$\begin{aligned} \text{constrained input} \;&\longrightarrow\; \text{initial full-round prefix} \;\longrightarrow\; \text{post-prefix state } \boldsymbol{q}_{R_f}, \\ \boldsymbol{q}_{R_f} \;&\longrightarrow\; \text{partial rounds} \;\longrightarrow\; \text{final full rounds.} \end{aligned}$$

Thus "post-prefix" identifies a position in the permutation schedule. Let $\mathcal{K}_r(M)$, defined formally in Section 2.4, denote the space of directions that keep the active S-box coordinate constant for the next r partial rounds. When the variable-dependent direction of $\boldsymbol{q}_{R_f}$ belongs to $\mathcal{K}_r(M)$, the same post-prefix state is also the *trail-entry state* of the resulting finite partial-round trail. Poseidon uses this permutation in sponge and compression modes; the present paper studies the permutation-level CICO problem [7].

2.3 The CICO Problem

Definition 1 (CICO-k). *Let $1 \le k < t$. In the zero-constraint form used here, a CICO-k solution for $\pi : \mathbb{F}^t \to \mathbb{F}^t$ is a state $\boldsymbol{s}$ satisfying*

$$s_i = 0 \quad \textit{and} \quad \pi(\boldsymbol{s})_i = 0, \qquad 0 \le i < k.$$

The general CICO definition permits arbitrary constrained coordinates and values. Our constructions use zeros in the first k coordinates, so we adopt that convention throughout. A generic permutation requires p^k evaluations to find such a state.

2.4 Finite Partial-Round Trails

During partial rounds we separate the state into a constant part and a part depending on attack variables. Round-constant addition affects only the constant part. If coordinate 0 of the variable-dependent part is zero, the partial S-box acts on a constant and does not increase degree in the attack variables.

For $r \geq 0$, define

$$\mathcal{K}_r(M) = \bigcap_{j=0}^{r-1} \ker(e_0^T M^j), \qquad \mathcal{K}_0(M) = \mathbb{F}^t. \tag{2}$$

This is the space $\mathcal{S}^{(r)}$ from Equation (1) of the Poseidon specification, written with zero-based coordinate indices [7, Section 2.3]. For $r \geq 1$, equivalently,

$$\mathcal{K}_r(M) = \operatorname{span}\{e_0, M^T e_0, \ldots, (M^T)^{r-1} e_0\}^{\perp}.$$

This is the orthogonal-Krylov subspace construction used by Beyne et al. to linearize partial rounds of HadesMiMC [5, Lemma 1].

Lemma 1 (Finite-trail space). *If the variable-dependent part entering the first partial round is $X\boldsymbol{a}$ with $\boldsymbol{a} \in \mathcal{K}_r(M)$, then the first r partial S-boxes act only on constants. Immediately before the matrix multiplication of partial round j, the variable-dependent direction is $M^j\boldsymbol{a}$ for $0 \leq j < r$. Moreover,*

$$\dim \mathcal{K}_r(M) \geq t - r.$$

Proof. At partial round j, the active coordinate has variable part $Xe_0^T M^j \boldsymbol{a} = 0$. The S-box therefore changes only the constant part, after which the linear layer advances the direction from $M^j\boldsymbol{a}$ to $M^{j+1}\boldsymbol{a}$. The dimension bound follows because (2) intersects at most r hyperplanes in a t-dimensional space.

Proposition 1 (Li–Wang block criterion [12, Proposition 1]). *Write*

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A \in \mathbb{F}, \quad B \in \mathbb{F}^{1\times(t-1)}, \quad C \in \mathbb{F}^{(t-1)\times 1}, \quad D \in \mathbb{F}^{(t-1)\times(t-1)}.$$

If $\boldsymbol{a} = (0, \boldsymbol{z})^T$, then

$$\boldsymbol{a} \in \mathcal{K}_r(M) \quad \Longleftrightarrow \quad BD^j\boldsymbol{z} = 0 \quad (0 \leq j \leq r-2), \tag{3}$$

with an empty condition when $r = 1$.

Proof. Starting from $(0, \boldsymbol{z})^T$, one linearized partial round produces $(B\boldsymbol{z}, D\boldsymbol{z})^T$. Requiring the next active coordinate to remain constant gives $B\boldsymbol{z} = 0$; iteration gives (3).

Define the observability matrix

$$\mathcal{O}_\ell = \begin{pmatrix} B \\ BD \\ \vdots \\ BD^{\ell-1} \end{pmatrix} \in \mathbb{F}^{\ell \times (t-1)}.$$

We say that M has *maximal partial-round rank growth* if

$$\operatorname{rank} \mathcal{O}_\ell = \ell \qquad (1 \le \ell \le t-1).$$

Then rank–nullity and (3) give

$$\dim \mathcal{K}_r(M) = t - r \quad (0 \le r \le t), \qquad \mathcal{K}_t(M) = \{0\}. \tag{4}$$

Consequently $t-1$ is the longest possible nonzero finite trail. This is the exact rank condition underlying the general-position bound of Li and Wang [12, Theorem 1]. For $t = 16$,

$$\dim \mathcal{K}_{14}(M) = 2, \qquad \dim \mathcal{K}_{15}(M) = 1, \qquad \mathcal{K}_{16}(M) = \{0\}.$$

2.5 Related Algebraic Attacks

Early analyses showed that particular choices of the HADES linear layer can produce invariant subspace trails through arbitrarily many partial rounds. Beyne et al. and Keller and Rosemarin used such weak linear layers to derive integral properties and preimage attacks for HadesMiMC and Starkad instances [5,11]. Grassi, Rechberger, and Schofnegger subsequently gave general criteria for excluding infinitely long invariant and iterative subspace trails [10]. Those criteria do not remove the finite trails used here: our concrete matrices pass the corresponding checks, while the construction enters a transient trail of bounded length.

Bariant, Bouvier, Leurent, and Perrin introduced a two-stage parametrization that removes the degree contribution of the first two Poseidon full rounds and reduces the remaining CICO-1 problem to univariate root finding [4]. Ashur, Buschman, and Mahzoun revisited the HADES security analysis and quantified Gröbner-basis attacks against Poseidon and Poseidon2 [1]. Grassi, Koschatko, and Rechberger later refined Gröbner-basis attacks that use finite subspace trails to linearize partial rounds of fixed Poseidon, Poseidon2, and Neptune instances [8]. Subsequent work solved reduced-round Poseidon and Poseidon2 bounty instances using resultants and univariate root finding [3]. Sanso and Vitto developed Graeffe-based root finding over NTT-friendly fields [15]; Zhao et al. subsequently improved this approach and established lower bounds for the memory-access cost of NTT computations [18]. Bak et al. developed the CheapLunch framework for multiple-output CICO systems [2].

Two recent works are particularly close in objective but distinct in mechanism. Merz and Rodríguez García exploit the prescribed tensor structure of

the Poseidon2 linear layers to improve round skipping for fixed Poseidon2 and Poseidon2b parameters [14]. Top Gun introduces degree annihilation: algebraic control equations cancel dominant degree contributions of rounds that remain in the model, and the technique is combined with the classical two-full-round skip [16].

Our solving stage uses these prior ideas: fix the control, form a univariate output polynomial, find its base-field roots, and test the remaining constrained outputs. We make no novelty claim for that stage. The distinct contribution is a Poseidon parameter-selection construction: after the constants are fixed, the action of M is prescribed on selected independent vectors so that a constrained-input family crosses *four* initial full rounds and enters the unavoidable finite space $\mathcal{K}_{14}(M)$. The earlier works do not construct such a round-constant-dependent MDS matrix.

3 The CICO-2 Construction

Our target Poseidon instance will be $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$,. The construction starts from a two-parameter family (x, w) immediately after the first full-round S-box. This family is chosen so that reversing the S-box automatically gives inputs whose first two coordinates are zero. We denote the resulting CICO-2 linear layer by M_2. It is constructed so that, at each subsequent full S-box in the initial prefix, constant terms and parameter-dependent terms occupy disjoint coordinate groups. Componentwise cubing can then raise the scalar functions of (x, w) without creating mixed terms or new vector directions.

After the four initial full rounds, the change of variable $X = x^{27}$ makes the state linear in X. The control w selects a direction in a two-dimensional space, and the remaining freedom in the linear layer is used to place that entire space inside $\mathcal{K}_{14}(M_2)$. The next fourteen partial S-boxes therefore act only on constants. What remains is a low-degree CICO-2 system: for each fixed control, the two constrained outputs are univariate polynomials in X.

3.1 Support-Separated Exponentiation

The construction repeatedly separates the coordinates carrying constants from those carrying family variables. Its goal is twofold. First, disjoint supports prevent cubing from producing mixed constant–variable terms. Second, by making two variable directions proportional within each of two coordinate groups, their linear combination has only one scalar coefficient per group. Cubing then changes those two coefficients but does not create new vector directions. The following lemma packages this step.

Lemma 2 (Two-group separation). *Let G, H, C partition $\{0, \dots, t-1\}$. Take $\boldsymbol{e}, \boldsymbol{v}_0, \boldsymbol{v}_1 \in \mathbb{F}^t$, and assume that $\boldsymbol{e}$ is supported on C, meaning that $e_i = 0$ for every $i \notin C$, and that $\boldsymbol{v}_0, \boldsymbol{v}_1$ vanish on C, meaning that $(v_0)_i = (v_1)_i = 0$ for*

every $i \in C$. Let $\lambda_g, \lambda_h \in \mathbb{F}$ satisfy

$$(v_1)_i = \begin{cases} \lambda_g (v_0)_i, & i \in G, \\ \lambda_h (v_0)_i, & i \in H. \end{cases}$$

Thus λ_g and λ_h are the proportionality ratios between the two directions on groups G and H, respectively. In the applications below they are chosen distinct; if they were equal, the two groups could be merged. Define $\boldsymbol{z}_g, \boldsymbol{z}_h \in \mathbb{F}^t$ coordinatewise by

$$(z_g)_i = \begin{cases} (v_0)_i^3, & i \in G, \\ 0, & i \notin G, \end{cases} \qquad (z_h)_i = \begin{cases} (v_0)_i^3, & i \in H, \\ 0, & i \notin H. \end{cases}$$

Because $\boldsymbol{v}_0$ vanishes on C, this gives $\boldsymbol{v}_0^{\circ 3} = \boldsymbol{z}_g + \boldsymbol{z}_h$.

Fix an integer $s \geq 0$. For any $f, g \in \mathbb{F}[W_1, \ldots, W_s]$ and any additional indeterminate Z, the following identity holds in $\mathbb{F}[W_1, \ldots, W_s, Z]^t$:

$$\begin{aligned} &(\boldsymbol{e} + Z(f\boldsymbol{v}_0 + g\boldsymbol{v}_1))^{\circ 3} \\ &\quad = \boldsymbol{e}^{\circ 3} + Z^3 \left((f + \lambda_g\, g)^3 \boldsymbol{z}_g + (f + \lambda_h\, g)^3 \boldsymbol{z}_h \right). \end{aligned} \tag{5}$$

Here f and g are scalar-valued polynomials in the control variables $W_1, \ldots, W_s$. When $s = 0$, they are simply scalars in $\mathbb{F}$. Since the displayed equality is a polynomial identity, it remains valid after substituting polynomials for any of the indeterminates, including Z.

Proof. It suffices to verify equality in each coordinate. Let $i \in C$. The vanishing assumptions give $(v_0)_i = (v_1)_i = 0$, while the definitions of $\boldsymbol{z}_g$ and $\boldsymbol{z}_h$ give $(z_g)_i = (z_h)_i = 0$. Hence coordinate i of both sides of (5) is e_i^3.

Now let $i \in G$. Since $\boldsymbol{e}$ is supported on C, we have $e_i = 0$. Moreover, $(v_1)_i = \lambda_g (v_0)_i$, so coordinate i of the left-hand side is

$$\begin{aligned} \left(Z\big(f \cdot (v_0)_i + g \cdot (v_1)_i\big) \right)^3 &= Z^3 \left(f + \lambda_g\, g\right)^3 (v_0)_i^3 \\ &= Z^3 \left(f + \lambda_g\, g\right)^3 (z_g)_i. \end{aligned}$$

Because $(z_h)_i = 0$, this is also coordinate i of the right-hand side. The same calculation for $i \in H$, with λ_h and $\boldsymbol{z}_h$ in place of λ_g and $\boldsymbol{z}_g$, proves equality on H. Since C, G, H partition all coordinates, the vector identity follows.

Here Z is a placeholder for the monomial carrying the dependence on the root parameter x. In the three successive applications below, we substitute $Z = x$, $Z = x^3$, and $Z = x^9$. Each S-box cubing replaces the current factor Z by Z^3, producing the sequence x^3, x^9, x^{27}. The polynomials f, g carry only the dependence on the control parameters, while λ_g, λ_h are fixed construction constants.

3.2 Four-Full-Round Identity

The initial full-round prefix is $R_3^F \circ R_2^F \circ R_1^F \circ R_0^F$. In this subsection, $j \in \{0,1,2,3\}$ denotes a full-round index: $\boldsymbol{c}^{(j)}$ is the constant added in round j, and, for $j \geq 1$, $\boldsymbol{\ell}_j$ denotes the input to the S-box of round j. We begin inside round 0, after its S-box and before its matrix multiplication, and denote this intermediate state by $\boldsymbol{q}_0$. Here x is the *root variable*, while w is the *control variable*. Fixing w selects a one-parameter fiber on which the output constraints will be solved in the reparametrized root variable $X = x^{27}$. We set

$$\boldsymbol{q}_0(x,w) = \boldsymbol{d}_c + x(\boldsymbol{d}_x + w\boldsymbol{d}_w). \tag{6}$$

Thus $\boldsymbol{d}_c$ is the constant term of the family, $\boldsymbol{d}_x$ is its root-variable direction, and $\boldsymbol{d}_w$ is the direction modulated by the control w. The original permutation input is recovered uniquely as

$$\boldsymbol{s}^{(0)}(x,w) = \sqrt[3]{\boldsymbol{q}_0(x,w)} - \boldsymbol{c}^{(0)}. \tag{7}$$

To impose $s_0^{(0)} = s_1^{(0)} = 0$ identically, we require

$$(d_x)_i = (d_w)_i = 0, \qquad (d_c)_i = (c_i^{(0)})^3, \qquad i \in \{0,1\}. \tag{8}$$

We use Lemma 2 for all three support-separated S-boxes. For each $j \in \{1,2,3\}$, choose coordinate groups G_j, H_j, C_j that partition the state coordinates, and a constant part $\boldsymbol{e}^{(j)}$ supported on C_j. Take $C_3 = \varnothing$, which forces $\boldsymbol{e}^{(3)} = 0$, and define

$$\boldsymbol{\varepsilon}^{(j)} = (\boldsymbol{e}^{(j)})^{\circ 3}.$$

Choose directions $\boldsymbol{v}_0^{(j)}, \boldsymbol{v}_1^{(j)} \in \mathbb{F}^t$ that vanish on C_j, and distinct ratios $\lambda_g^{(j)}, \lambda_h^{(j)} \in \mathbb{F}$ satisfying

$$(v_1^{(j)})_i = \begin{cases} \lambda_g^{(j)} (v_0^{(j)})_i, & i \in G_j, \\ \lambda_h^{(j)} (v_0^{(j)})_i, & i \in H_j. \end{cases} \tag{9}$$

Define the corresponding group-supported cube components by

$$(\boldsymbol{v}_0^{(j)})^{\circ 3} = \boldsymbol{z}_g^{(j)} + \boldsymbol{z}_h^{(j)}, \qquad \operatorname{supp}(\boldsymbol{z}_g^{(j)}) \subseteq G_j, \quad \operatorname{supp}(\boldsymbol{z}_h^{(j)}) \subseteq H_j. \tag{10}$$

The scalar coefficient polynomials follow the same stage numbering. Set $\gamma_g^{(0)}(w) = 1$ and $\gamma_h^{(0)}(w) = w$, and, for $1 \leq j \leq 3$, define

$$\begin{aligned} \gamma_g^{(j)} &= \left(\gamma_g^{(j-1)} + \lambda_g^{(j)} \gamma_h^{(j-1)}\right)^3, \\ \gamma_h^{(j)} &= \left(\gamma_g^{(j-1)} + \lambda_h^{(j)} \gamma_h^{(j-1)}\right)^3. \end{aligned} \tag{11}$$

Thus stage j uses the directions, ratios, and group components carrying the same superscript (j).

For stage 1, prescribe

$$M_2 \boldsymbol{d}_x = \boldsymbol{v}_0^{(1)}, \qquad M_2 \boldsymbol{d}_w = \boldsymbol{v}_1^{(1)}, \qquad M_2 \boldsymbol{d}_c = \boldsymbol{e}^{(1)} - \boldsymbol{c}^{(1)}. \tag{12}$$

The input to the S-box of full round 1 is therefore

$$\boldsymbol{\ell}_1 = \boldsymbol{e}^{(1)} + x\left(\gamma_g^{(0)}\boldsymbol{v}_0^{(1)} + \gamma_h^{(0)}\boldsymbol{v}_1^{(1)}\right).$$

Lemma 2 gives

$$\boldsymbol{\ell}_1^{\circ 3} = \boldsymbol{\varepsilon}^{(1)} + x^3\left(\gamma_g^{(1)}\boldsymbol{z}_g^{(1)} + \gamma_h^{(1)}\boldsymbol{z}_h^{(1)}\right). \tag{13}$$

For stage 2, prescribe

$$M_2\boldsymbol{\varepsilon}^{(1)} = \boldsymbol{e}^{(2)} - \boldsymbol{c}^{(2)}, \qquad M_2\boldsymbol{z}_g^{(1)} = \boldsymbol{v}_0^{(2)}, \qquad M_2\boldsymbol{z}_h^{(1)} = \boldsymbol{v}_1^{(2)}. \tag{14}$$

The input to the S-box of full round 2 is

$$\boldsymbol{\ell}_2 = \boldsymbol{e}^{(2)} + x^3\left(\gamma_g^{(1)}\boldsymbol{v}_0^{(2)} + \gamma_h^{(1)}\boldsymbol{v}_1^{(2)}\right).$$

A second application of Lemma 2 gives

$$\boldsymbol{\ell}_2^{\circ 3} = \boldsymbol{\varepsilon}^{(2)} + x^9\left(\gamma_g^{(2)}\boldsymbol{z}_g^{(2)} + \gamma_h^{(2)}\boldsymbol{z}_h^{(2)}\right). \tag{15}$$

For stage 3, prescribe

$$M_2\boldsymbol{\varepsilon}^{(2)} = -\boldsymbol{c}^{(3)}, \qquad M_2\boldsymbol{z}_g^{(2)} = \boldsymbol{v}_0^{(3)}, \qquad M_2\boldsymbol{z}_h^{(2)} = \boldsymbol{v}_1^{(3)}. \tag{16}$$

The first equality cancels the constant term before the S-box of full round 3, whose input is therefore

$$\boldsymbol{\ell}_3 = x^9\left(\gamma_g^{(2)}\boldsymbol{v}_0^{(3)} + \gamma_h^{(2)}\boldsymbol{v}_1^{(3)}\right).$$

After cubing, the group components are $x^{27}(\gamma_g^{(3)}\boldsymbol{z}_g^{(3)} + \gamma_h^{(3)}\boldsymbol{z}_h^{(3)})$. Label the two final groups so that coordinate 0 lies in G_3. Consequently $\boldsymbol{z}_h^{(3)}$, which is supported on H_3, has zero first coordinate. Set $\boldsymbol{u}_1 = \boldsymbol{z}_h^{(3)}$, choose $\boldsymbol{u}_2$ with zero first coordinate, and prescribe

$$M_2(\boldsymbol{z}_g^{(3)} + \boldsymbol{z}_h^{(3)}) = \boldsymbol{u}_1, \qquad M_2\boldsymbol{z}_h^{(3)} = \boldsymbol{u}_2. \tag{17}$$

Proposition 2 (Four-round absorption identity). *Under* (12), (14), (16), *and* (17), *the post-prefix state after the fourth full-round matrix is*

$$\boldsymbol{q}_4(X, w) = X\left(\gamma_g^{(3)}(w)\boldsymbol{u}_1 + (\gamma_h^{(3)}(w) - \gamma_g^{(3)}(w))\boldsymbol{u}_2\right), \qquad X = x^{27}. \tag{18}$$

For each fixed w, this state is linear in X. The vector multiplying X is its post-prefix direction. *After the trail completion of Section 3.3, this direction lies in $\mathcal{K}_{14}(M_2)$, so the post-prefix state is also the entry state of the subsequent finite partial-round trail.*

Proof. Equations (13) and (14) give $\boldsymbol{\ell}_2$. Equation (15) and (16) give $\boldsymbol{\ell}_3$, with its constant term cancelled. Cubing $\boldsymbol{\ell}_3$ and applying Lemma 2 gives

$$x^{27}(\gamma_g^{(3)}\boldsymbol{z}_g^{(3)} + \gamma_h^{(3)}\boldsymbol{z}_h^{(3)}).$$

Since (17) implies $M_2\boldsymbol{z}_g^{(3)} = \boldsymbol{u}_1 - \boldsymbol{u}_2$ and $M_2\boldsymbol{z}_h^{(3)} = \boldsymbol{u}_2$, the last matrix multiplication gives (18).

Because cubing is bijective on the concrete field, so is $x \mapsto x^{27}$. Every $X \in \mathbb{F}_p$ therefore determines a unique x by three successive cube roots, after which (7) determines a unique constrained input.

3.3 Matrix Construction and Trail Completion

The prefix construction fixes the action of M_2 on eleven independent vectors and, in particular, gives $M_2\boldsymbol{u}_1 = \boldsymbol{u}_2$. Our next goal is to extend this relation to a Krylov chain

$$M_2\boldsymbol{u}_i = \boldsymbol{u}_{i+1} \qquad (1 \le i < 15)$$

whose vectors all have zero in coordinate 0. This immediately keeps both $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ inactive for fourteen partial rounds: the successive directions from $\boldsymbol{u}_1$ are $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{14}$, while those from $\boldsymbol{u}_2$ are $\boldsymbol{u}_2, \ldots, \boldsymbol{u}_{15}$.

Choose $\boldsymbol{u}_3, \ldots, \boldsymbol{u}_6$, all with zero first coordinate, so that the following sixteen vectors form a basis of $\mathbb{F}^{16}$:

$$\big(\boldsymbol{d}_x, \boldsymbol{d}_w, \boldsymbol{d}_c, \boldsymbol{\varepsilon}^{(1)}, \boldsymbol{z}_g^{(1)}, \boldsymbol{z}_h^{(1)}, \boldsymbol{\varepsilon}^{(2)}, \boldsymbol{z}_g^{(2)}, \boldsymbol{z}_h^{(2)}, \boldsymbol{z}_g^{(3)} + \boldsymbol{z}_h^{(3)}, \boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_6\big).$$

The prefix prescribes the images of the first eleven basis vectors, ending with $\boldsymbol{u}_1 \mapsto \boldsymbol{u}_2$. To extend the chain, we also prescribe $\boldsymbol{u}_i \mapsto \boldsymbol{u}_{i+1}$ for $2 \le i \le 5$. Only $\boldsymbol{u}_6$ has not been assigned an image: the prefix does not constrain it, so we can choose it to extend the inactive trail.

Before making that choice, define $M_{2,0}$ by keeping the first fifteen assignments and setting only the still-unassigned image of $\boldsymbol{u}_6$ to zero. Its action on the entire basis above is

$$\begin{aligned}
M_{2,0}\boldsymbol{d}_x &= \boldsymbol{v}_0^{(1)}, & M_{2,0}\boldsymbol{d}_w &= \boldsymbol{v}_1^{(1)},\\
M_{2,0}\boldsymbol{d}_c &= \boldsymbol{e}^{(1)} - \boldsymbol{c}^{(1)}, & M_{2,0}\boldsymbol{\varepsilon}^{(1)} &= \boldsymbol{e}^{(2)} - \boldsymbol{c}^{(2)},\\
M_{2,0}\boldsymbol{z}_g^{(1)} &= \boldsymbol{v}_0^{(2)}, & M_{2,0}\boldsymbol{z}_h^{(1)} &= \boldsymbol{v}_1^{(2)},\\
M_{2,0}\boldsymbol{\varepsilon}^{(2)} &= -\boldsymbol{c}^{(3)}, & M_{2,0}\boldsymbol{z}_g^{(2)} &= \boldsymbol{v}_0^{(3)},\\
M_{2,0}\boldsymbol{z}_h^{(2)} &= \boldsymbol{v}_1^{(3)}, & M_{2,0}(\boldsymbol{z}_g^{(3)} + \boldsymbol{z}_h^{(3)}) &= \boldsymbol{u}_1,\\
M_{2,0}\boldsymbol{u}_i &= \boldsymbol{u}_{i+1} \quad (1 \le i \le 5), & M_{2,0}\boldsymbol{u}_6 &= 0.
\end{aligned}$$

Each equality pairs one basis vector with its image. Because the vectors on the left form a basis, these assignments determine one and only one linear map. Every vector appearing here has already been chosen, so $M_{2,0}$ does not depend on $\boldsymbol{u}_7$ or on the final matrix M_2.

There are nine positions left in the desired chain, namely $\boldsymbol{u}_7, \ldots, \boldsymbol{u}_{15}$. We use the name $\boldsymbol{u}_7$ for the vector that will be assigned as the image of $\boldsymbol{u}_6$, and choose it by requiring

$$e_0^T M_{2,0}^j \boldsymbol{u}_7 = 0 \qquad (0 \le j \le 8).$$

The upper bound is 8 because, under the final map M_2, $\boldsymbol{u}_7 = M_2^0 \boldsymbol{u}_7$ and $\boldsymbol{u}_{15} = M_2^8 \boldsymbol{u}_7$. We impose the same nine exponents on the already known map $M_{2,0}$; the induction below shows that these conditions also control the orbit under the final map M_2. They are nine homogeneous linear equations in the sixteen coordinates of $\boldsymbol{u}_7$, so their solution space has dimension at least seven.

After selecting such a vector, define M_2 as the unique linear map that agrees with $M_{2,0}$ on the first fifteen basis vectors displayed above and instead assigns

$$M_2 \boldsymbol{u}_6 = \boldsymbol{u}_7.$$

Although these equations use $M_{2,0}$, they also control iteration by the map M_2. The two maps differ only in the image assigned to $\boldsymbol{u}_6$; consequently, their difference sends every vector to a scalar multiple of $\boldsymbol{u}_7$. An induction on j therefore gives

$$M_2^j \boldsymbol{u}_7 \in \operatorname{span}\{\boldsymbol{u}_7, M_{2,0}\boldsymbol{u}_7, \ldots, M_{2,0}^j \boldsymbol{u}_7\}.$$

Every vector on the right has zero first coordinate for $0 \le j \le 8$. Thus the actual orbit $\boldsymbol{u}_7, \boldsymbol{u}_8, \ldots, \boldsymbol{u}_{15}$, where $\boldsymbol{u}_{i+1} = M_2 \boldsymbol{u}_i$, has the required zero coordinate.

We can now verify membership in $\mathcal{K}_{14}(M_2)$ directly. By definition, a vector $\boldsymbol{a}$ belongs to this space precisely when

$$e_0^T M_2^j \boldsymbol{a} = 0 \qquad (0 \le j \le 13).$$

The vectors $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ were chosen with zero first coordinate, as were $\boldsymbol{u}_3, \ldots, \boldsymbol{u}_6$; the preceding argument establishes the same property for $\boldsymbol{u}_7, \ldots, \boldsymbol{u}_{15}$. Moreover, the chain relation gives, for every $0 \le j \le 13$,

$$M_2^j \boldsymbol{u}_1 = \boldsymbol{u}_{j+1}, \qquad M_2^j \boldsymbol{u}_2 = \boldsymbol{u}_{j+2}.$$

The first orbit therefore visits $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{14}$, and the second visits $\boldsymbol{u}_2, \ldots, \boldsymbol{u}_{15}$; every visited vector has zero first coordinate. Hence both $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ belong to $\mathcal{K}_{14}(M_2)$. Since $\mathcal{K}_{14}(M_2)$ is a vector space, it contains every linear combination of them, proving

$$\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2\} \subseteq \mathcal{K}_{14}(M_2). \tag{19}$$

The same argument for one additional power shows that $\boldsymbol{u}_1 \in \mathcal{K}_{15}(M_2)$, since its first fifteen orbit vectors are $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{15}$. If M_2 has maximal rank growth and $\boldsymbol{u}_1, \boldsymbol{u}_2$ are independent, then both sides have dimension two and equality holds.

From (18), after j partial rounds the variable-dependent direction is

$$\boldsymbol{\delta}_j(w) = \gamma_g^{(3)}(w)\boldsymbol{u}_{j+1} + (\gamma_h^{(3)}(w) - \gamma_g^{(3)}(w))\boldsymbol{u}_{j+2}.$$

It has zero active coordinate for $0 \le j \le 13$; therefore fourteen partial S-boxes act only on constants. We call a control *live* when the post-prefix direction is nonzero and $\gamma_h^{(3)}(w) \neq \gamma_g^{(3)}(w)$. If a control $w_\star$ satisfies $\gamma_h^{(3)}(w_\star) = \gamma_g^{(3)}(w_\star) \neq 0$, the post-prefix direction collapses to $\boldsymbol{u}_1 \in \mathcal{K}_{15}(M_2)$, and the fifteenth partial S-box is also inactive.

3.4 Concrete CICO-2 Solution

The instance uses

$$p = 2130706433, \qquad t = 16, \qquad \alpha = 3, \qquad R_F = 8, \qquad R_P = 20.$$

The schedule and constants come from the Poseidon reference implementation.[1] We used:

```
sage generate_params_poseidon.sage   1 0 31 16 3 128 0x7f000001
sage generate_parameters_grain.sage  1 0 31 16 8 20 0x7f000001 .
```

The construction treats the resulting $448 = 16(8 + 20)$ round constants as fixed data and replaces only the generated matrix.

The three support partitions, in the order used in Section 3.2, are

$$\begin{aligned}(G_1, H_1, C_1) &= \big(\{0, \dots, 6\}, \{7, \dots, 13\}, \{14, 15\}\big),\\ (G_2, H_2, C_2) &= \big(\{0, \dots, 7\}, \{8, \dots, 13\}, \{14, 15\}\big),\\ (G_3, H_3) &= \big(\{0, \dots, 8\}, \{9, \dots, 15\}\big).\end{aligned}$$

Together with M_2, the constants, and the vectors and ratios in Appendix A, these partitions determine every intermediate vector in (12)–(17).

Direct substitution verifies that the input family (6) has its first two coordinates identically zero and that all prescribed images in (12)–(17) hold over $\mathbb{F}_p$. The four-round identity (18) therefore holds for the concrete data.

The two post-prefix directions $\boldsymbol{u}_1, \boldsymbol{u}_2$ are independent. For the constructed matrix, exact linear algebra gives

$$\big(\operatorname{rank} \mathcal{O}_1, \dots, \operatorname{rank} \mathcal{O}_{15}\big) = (1, 2, \dots, 15).$$

In particular, $\dim \mathcal{K}_{14}(M_2) = 2$, so the inclusion (19) is an equality. Thus a live control gives fourteen inactive partial S-boxes. Direct evaluation of (11) also gives the collapsed control

$$w_\star = 244601253, \qquad \gamma_h^{(3)}(w_\star) = \gamma_g^{(3)}(w_\star) \neq 0,$$

for which the post-prefix direction lies on $\operatorname{span}\{\boldsymbol{u}_1\} \subseteq \mathcal{K}_{15}(M_2)$, making one additional partial S-box inactive.

We verified the matrix properties by exact finite-field computation. Its determinant is

$$\det(M_2) = 437213552 \neq 0.$$

Every minor of sizes $1, \dots, 8$ in M_2, and every minor of sizes $1, \dots, 7$ in M_2^{-1}, is nonzero. Jacobi's complementary-minor identity covers sizes $9, \dots, 15$, and the determinant covers size 16. Thus all

$$\sum_{j=1}^{16} \binom{16}{j}^2 = \binom{32}{16} - 1 = 601080389$$

[1] `https://extgit.isec.tugraz.at/krypto/hadeshash`

nonempty square minors are nonzero. The observability ranks and the irreducibility of $\chi_{M_2^i}$ for every $1 \le i \le 64$ were checked by exact linear algebra and polynomial factorization.

For a fixed control w, let $F_{i,w}(X)$ denote output coordinate i obtained by continuing the permutation from $\boldsymbol{q}_4(X, w)$. At the control

$$w^\dagger = 1203195526,$$

symbolic evaluation of the remaining rounds gives two constrained-output polynomials of exact degree

$$3^{10} = 59049,$$

as verified by their nonzero leading terms. They have the common base-field root

$$X^\dagger = 1404166674.$$

This degree is the contribution of the six partial S-boxes that remain active after the fourteen-round trail and the four final full S-box layers; the direct layer-counting bound for a generic affine input would be 3^{28}. At the collapsed control $w_\star$, only five partial S-boxes remain active, giving the smaller bound 3^9.

Three successive cube roots of $X^\dagger$ give $x^\dagger = 935462835$. Applying (7) gives the input in Appendix A; exact evaluation of all twenty-eight rounds gives the printed output with its first two coordinates zero. The recovered input and output therefore form the reported CICO-2 solution.

Zero-Test Application The bounty's zero-test challenge encodes the $t = 16$ input as the eight coefficients of a degree-seven polynomial $P \in \mathbb{F}_{p^2}[Z]$. The first pair of compression outputs defines $a_0 \in \mathbb{F}_{p^2}$, and a solution must satisfy $P(a_0) = 0$ [6]. Applying an analogous construction with $R_F = 6$, we obtained a solution to this challenge. This is separate from the full-round $R_F = 8, R_P = 20$ CICO-2 solution above.

4 The CICO-3 Construction

The CICO-3 construction follows the same sequence as CICO-2: constrain the input after the first S-box, propagate group coefficients through four full rounds, construct the matrix from prescribed images, and complete a Krylov chain. It targets the same Poseidon parameters,

$$(t, \alpha, R_F, R_P) = (16, 3, 8, 20), \qquad R_f = 4.$$

The two differences are that there are two controls and three support groups.

4.1 Three-Group Four-Round Identity

At each of the three remaining full S-boxes, a 3×3 ratio matrix updates the scalar coefficients on disjoint support groups. After the last S-box, a change of

basis lets the group vectors begin the partial-round trail without violating the MDS branch-number condition.

Here x is the root variable and w, z are the two control variables. Fixing (w, z) leaves a one-parameter family in

$$X = x^{27}.$$

As in CICO-2, we begin immediately after the S-box of full round 0:

$$\boldsymbol{q}_0(x, w, z) = \boldsymbol{d}_c + x(\boldsymbol{d}_x + w\boldsymbol{d}_w + z\boldsymbol{d}_z). \tag{20}$$

The corresponding permutation input is recovered coordinatewise as

$$\boldsymbol{s}^{(0)}(x, w, z) = \sqrt[3]{\boldsymbol{q}_0(x, w, z)} - \boldsymbol{c}^{(0)}.$$

Consequently, its first three coordinates vanish identically when

$$(d_x)_i = (d_w)_i = (d_z)_i = 0, \qquad (d_c)_i = (c_i^{(0)})^3, \qquad 0 \le i < 3. \tag{21}$$

Let $s \in \{1, 2, 3\}$ index the remaining full S-boxes, and let $g, k \in \{1, 2, 3\}$ index support groups and incoming directions, respectively.

For stage s, write the three variable-support groups as $G_1^{(s)}, G_2^{(s)}, G_3^{(s)}$. At the first two stages, the remaining coordinates form a constant-support set $C^{(s)}$; the third stage has no constant part. The concrete partitions are

$$\begin{aligned}
(G_1^{(1)}, G_2^{(1)}, G_3^{(1)}; C^{(1)}) &= (\{0, \dots, 4\}, \{5, \dots, 9\}, \{10, \dots, 13\}; \{14, 15\}),\\
(G_1^{(2)}, G_2^{(2)}, G_3^{(2)}; C^{(2)}) &= (\{0, \dots, 4\}, \{5, \dots, 9\}, \{10, 11, 14, 15\}; \{12, 13\}),\\
(G_1^{(3)}, G_2^{(3)}, G_3^{(3)}) &= (\{0, 1, 2\}, \{3, \dots, 8\}, \{9, \dots, 15\}).
\end{aligned} \tag{22}$$

The semicolon separates the three variable groups from the constant group. In particular, the final variable groups have sizes $3, 6, 7$.

Collect the group–direction ratios into

$$\Lambda^{(s)} = \left(\lambda_{g,k}^{(s)}\right)_{g,k=1}^{3} \in \mathrm{GL}_3(\mathbb{F}).$$

Row g corresponds to group $G_g^{(s)}$ and column k to direction $\boldsymbol{v}_k^{(s)}$. Choose $\boldsymbol{r}_g^{(s)}$ with support exactly $G_g^{(s)}$, and set

$$\boldsymbol{v}_k^{(s)} = \sum_{g=1}^{3} \lambda_{g,k}^{(s)} \boldsymbol{r}_g^{(s)} \qquad (1 \le k \le 3). \tag{23}$$

These directions are proportional within each group. Write $\boldsymbol{z}_g^{(s)} = (\boldsymbol{r}_g^{(s)})^{\circ 3}$. If $\boldsymbol{e}^{(s)}$ is supported on the disjoint constant set $C^{(s)}$, and $\boldsymbol{\varepsilon}^{(s)} = (\boldsymbol{e}^{(s)})^{\circ 3}$, then for any coefficient column $\boldsymbol{a} = (a_1, a_2, a_3)^T$ and scalar ξ,

$$\left(\boldsymbol{e}^{(s)} + \xi \sum_{k=1}^{3} a_k \boldsymbol{v}_k^{(s)}\right)^{\circ 3} = \boldsymbol{\varepsilon}^{(s)} + \xi^3 \sum_{g=1}^{3} (\Lambda^{(s)}\boldsymbol{a})_g^3 \boldsymbol{z}_g^{(s)}. \tag{24}$$

This is the three-group version of Lemma 2. Disjoint supports are the reason no mixed terms appear.

Invertibility of $\Lambda^{(s)}$ prevents a linear update from collapsing the coefficient directions. All three stages use the recurrence

$$\boldsymbol{\gamma}^{(0)}(w,z) = (1,w,z)^T, \qquad \boldsymbol{\gamma}^{(s)} = \left(\Lambda^{(s)}\boldsymbol{\gamma}^{(s-1)}\right)^{\circ 3} \quad (1 \le s \le 3). \tag{25}$$

Thus $\boldsymbol{\gamma}^{(s)}$ is the column of three scalar coefficients immediately after the S-box at stage s.

For the three support-separated transitions, prescribe

$$\begin{aligned} &M_3\boldsymbol{d}_x = \boldsymbol{v}_1^{(1)}, && M_3\boldsymbol{d}_w = \boldsymbol{v}_2^{(1)}, && M_3\boldsymbol{d}_z = \boldsymbol{v}_3^{(1)}, \\ &M_3\boldsymbol{d}_c = \boldsymbol{e}^{(1)} - \boldsymbol{c}^{(1)}, && M_3\boldsymbol{\varepsilon}^{(1)} = \boldsymbol{e}^{(2)} - \boldsymbol{c}^{(2)}, && M_3\boldsymbol{\varepsilon}^{(2)} = -\boldsymbol{c}^{(3)}, \\ &M_3\boldsymbol{z}_k^{(s)} = \boldsymbol{v}_k^{(s+1)} && (1 \le k \le 3,\ s = 1,2). \end{aligned} \tag{26}$$

Here $\boldsymbol{e}^{(s)}$ is supported on $C^{(s)}$ for $s = 1,2$, and $\boldsymbol{e}^{(3)} = \boldsymbol{\varepsilon}^{(3)} = 0$. Induction using (24) gives the S-box inputs and outputs

$$\begin{aligned} \boldsymbol{\ell}_s &= \boldsymbol{e}^{(s)} + x^{3^{s-1}} \sum_{k=1}^{3} \gamma_k^{(s-1)} \boldsymbol{v}_k^{(s)}, \\ \boldsymbol{\ell}_s^{\circ 3} &= \boldsymbol{\varepsilon}^{(s)} + x^{3^s} \sum_{g=1}^{3} \gamma_g^{(s)} \boldsymbol{z}_g^{(s)} \qquad (1 \le s \le 3). \end{aligned}$$

In particular, cancellation of the round-3 constants gives

$$\boldsymbol{\ell}_3^{\circ 3} = x^{27} \sum_{g=1}^{3} \gamma_g^{(3)} \boldsymbol{z}_g^{(3)}. \tag{27}$$

The expanded stage-by-stage calculation is recorded in Appendix C.

It remains to pass this three-dimensional combination through the fourth matrix multiplication while starting the inactive matrix orbit. Each vector $\boldsymbol{z}_g^{(3)}$ is supported on its final group, of size 3, 6, or 7. It would be tempting to use these sparse vectors directly as the beginning of a matrix orbit. That is incompatible with an MDS matrix: for example, mapping the weight-3 vector $\boldsymbol{z}_1^{(3)}$ to any one of the other final group vectors would give input and output weights summing to at most $3 + 7 < 17$, contradicting the branch-number bound for $t = 16$.

We therefore replace the three sparse group vectors by another basis of their span. Choose an invertible matrix of the form

$$P = \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix},$$

where the three entries in the first row are nonzero and the lower-right 2×2 block is invertible, and define

$$[\,\boldsymbol{b}_0\ \boldsymbol{u}_1\ \boldsymbol{u}_2\,] = [\,\boldsymbol{z}_1^{(3)}\ \boldsymbol{z}_2^{(3)}\ \boldsymbol{z}_3^{(3)}\,]P^T. \tag{28}$$

The rows of P express $\boldsymbol{b}_0, \boldsymbol{u}_1, \boldsymbol{u}_2$ in the sparse group basis. Thus $\boldsymbol{b}_0$ is dense, while $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ are supported on $G_2^{(3)} \cup G_3^{(3)}$, which excludes active coordinate 0. The selected P and the ratio matrices are given in Appendix B.

This mixed basis lets the final prefix matrix multiplication both output inactive directions and begin their matrix orbit. Choose a further vector $\boldsymbol{u}_3$ with zero first coordinate and prescribe

$$M_3\boldsymbol{b}_0 = \boldsymbol{u}_1, \qquad M_3\boldsymbol{u}_1 = \boldsymbol{u}_2, \qquad M_3\boldsymbol{u}_2 = \boldsymbol{u}_3. \tag{29}$$

To express the state (27) in this new basis, define

$$\boldsymbol{h}(w,z) = (P^T)^{-1}\boldsymbol{\gamma}^{(3)}(w,z). \tag{30}$$

Indeed, (28) implies

$$\sum_{g=1}^{3} \gamma_g^{(3)} \boldsymbol{z}_g^{(3)} = h_0\boldsymbol{b}_0 + h_1\boldsymbol{u}_1 + h_2\boldsymbol{u}_2.$$

Proposition 3 (CICO-3 four-round identity). *Under* (26) *and* (29)*, the state after the fourth full-round matrix is*

$$\boldsymbol{q}_4(X,w,z) = X\big(h_0(w,z)\boldsymbol{u}_1 + h_1(w,z)\boldsymbol{u}_2 + h_2(w,z)\boldsymbol{u}_3\big), \qquad X = x^{27}. \tag{31}$$

The vector multiplying X is the post-prefix direction selected by the controls (w,z).

Proof. Equation (27) gives the state immediately before the fourth matrix multiplication. Equations (28) and (30) rewrite its three group components as

$$X(h_0\boldsymbol{b}_0 + h_1\boldsymbol{u}_1 + h_2\boldsymbol{u}_2).$$

Applying the three prescribed images in (29) gives (31).

4.2 Matrix Construction and Trail Completion

The prefix already gives $M_3\boldsymbol{u}_1 = \boldsymbol{u}_2$ and $M_3\boldsymbol{u}_2 = \boldsymbol{u}_3$. We want to continue this orbit through $\boldsymbol{u}_{15}$, with every vector having zero in coordinate 0. The resulting chain keeps every post-prefix direction in $\mathrm{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$ inactive for thirteen partial rounds.

The four prefix transitions prescribe fifteen independent domain–image pairs: four pairs in each of the first three transitions and three in the last. The vector $\boldsymbol{u}_3$ completes these fifteen domain vectors to a basis, so its image is the only one not prescribed by the prefix. Define $M_{3,0}$ by keeping all fifteen prefix assignments and setting only $M_{3,0}\boldsymbol{u}_3 = 0$. This map is independent of $\boldsymbol{u}_4$ and of the final matrix M_3.

The twelve positions still missing from the desired orbit are $\boldsymbol{u}_4, \ldots, \boldsymbol{u}_{15}$. We therefore choose $\boldsymbol{u}_4$ by requiring

$$e_0^T M_{3,0}^j \boldsymbol{u}_4 = 0 \qquad (0 \le j \le 11).$$

These twelve homogeneous linear equations in sixteen coordinates leave a solution space of dimension at least four.

After this choice, define the final map M_3 by replacing the zero image of $\boldsymbol{u}_3$ with $\boldsymbol{u}_4$. Hence $M_3\boldsymbol{u}_3 = \boldsymbol{u}_4$, and M_3 differs from $M_{3,0}$ only in that image. Their difference therefore always produces a scalar multiple of $\boldsymbol{u}_4$. Exactly as in the CICO-2 construction, induction gives

$$M_3^j \boldsymbol{u}_4 \in \operatorname{span}\{\boldsymbol{u}_4, M_{3,0}\boldsymbol{u}_4, \ldots, M_{3,0}^j \boldsymbol{u}_4\}.$$

For $0 \le j \le 11$, every vector on the right has zero first coordinate. Thus the actual orbit $\boldsymbol{u}_4, \ldots, \boldsymbol{u}_{15}$ has zero first coordinate as well. Since this property is imposed directly on $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$, we obtain

$$\operatorname{span}\{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\} \subseteq \mathcal{K}_{13}(M_3). \tag{32}$$

The final support groups are constrained by MDS. The vectors $\boldsymbol{z}_j^{(2)}$ are supported together on the fourteen coordinates outside $C^{(2)}$. For any final group $G_\ell^{(3)}$, take $\boldsymbol{\mu} = (\Lambda^{(3)})^{-1} e_\ell$, with e_ℓ the ℓ-th standard basis vector of $\mathbb{F}^3$, and set $\boldsymbol{v} = \sum_{j=1}^{3} \mu_j \boldsymbol{z}_j^{(2)}$. By (23) and (26), $M_3\boldsymbol{v} = \boldsymbol{r}_\ell^{(3)} \neq 0$, so the branch-number bound gives

$$17 \le \operatorname{wt}(\boldsymbol{v}) + \operatorname{wt}(M_3\boldsymbol{v}) \le 14 + |G_\ell^{(3)}|.$$

Every final group must therefore have size at least three. The chosen sizes $3, 6, 7$ meet this necessary condition; a group of size two would be incompatible with MDS. Appendix D states the general output-group bound and its proof.

4.3 Concrete CICO-3 Degree-Reduction Instance

We instantiate the construction over the same field, round schedule, and round constants as in Section 3.4. The matrix M_3, the ratio matrices, the basis change, and the input vectors are recorded in Appendix B. Direct substitution into (21) verifies that the first three input coordinates vanish identically. The prescribed domain–image relations also hold exactly over $\mathbb{F}_p$, so Proposition 3 gives the polynomial prefix identity (31) over $\mathbb{F}_p[w, z, x]$.

Exact linear algebra shows that the three post-prefix directions $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$ are independent and that M_3 has observability rank profile

$$\big(\operatorname{rank}\mathcal{O}_1, \ldots, \operatorname{rank}\mathcal{O}_{15}\big) = (1, 2, \ldots, 15).$$

Thus $\dim \mathcal{K}_{13}(M_3) = 3$, and the inclusion (32) is an equality. Every post-prefix direction selected by the controls therefore lies in $\mathcal{K}_{13}(M_3)$, so the first thirteen partial S-boxes are inactive in the univariate representation.

The constructed matrix orbit also satisfies

$$(u_i)_0 = 0 \ (1 \le i \le 15), \qquad (u_{16})_0 = 1756167053 \neq 0.$$

We also verified the matrix properties by exact finite-field computation. The determinant is $1802530722 \neq 0$. Minor checks through size eight in M_3 and

through size seven in M_3^{-1}, followed by Jacobi's complementary-minor identity, cover all 601080389 nonempty square minors and show that M_3 is MDS. Exact polynomial factorization further shows that $\chi_{M_3^i}$ is irreducible of degree 16 for every $1 \le i \le 64$.

For fixed controls (w, z), let $F_{i,w,z}(X)$ be output coordinate i obtained by continuing the permutation from (31). After the thirteen inactive partial rounds, seven active partial S-boxes and four final full S-box layers remain, so

$$\deg F_{i,w,z} \le 3^{7+4} = 3^{11} = 177147 < p.$$

It suffices to exhibit one control pair at which all three leading coefficients are nonzero. Exact propagation at $(w, z) = (0, 0)$ verifies this, so all three outputs have degree 3^{11} at that control.

These computations establish a CICO-3 degree-reduction instance with an MDS matrix and the three-dimensional trail-entry space $\mathcal{K}_{13}(M_3)$. We do not report a common root of the three constrained output polynomials, so this instance is not presented as a complete CICO-3 solution.

5 Generalization to CICO-k

This section isolates the two resources already visible in CICO-2 and CICO-3: the dimension of the post-prefix direction space and the number of independent images available when defining M. The results apply to the direct support-separated construction; they do not rule out different algebraic parametrizations.

5.1 Controls and Post-Prefix Direction Space

Let $\boldsymbol{w} = (w_1, \ldots, w_n)$ be n controls and replace (6) by

$$\boldsymbol{q}_0(x, \boldsymbol{w}) = \boldsymbol{d}_c + x \left(\boldsymbol{d}_x + \sum_{\ell=1}^{n} w_\ell \boldsymbol{d}_\ell \right). \tag{33}$$

The first k input coordinates vanish identically if

$$(d_x)_i = (d_\ell)_i = 0, \qquad (d_c)_i = (c_i^{(0)})^\alpha, \quad 0 \le i < k.$$

Suppose the R_f initial full rounds map this family to

$$\boldsymbol{q}_{R_f}(X, \boldsymbol{w}) = \boldsymbol{b}(\boldsymbol{w}) + X\boldsymbol{\delta}(\boldsymbol{w}).$$

The controls are held fixed when the remaining permutation is represented as a univariate polynomial in X. Consistently with Section 2.2, $\boldsymbol{q}_{R_f}$ is the post-prefix state and $\boldsymbol{\delta}(\boldsymbol{w})$ is its post-prefix direction.

Define the *post-prefix direction space*

$$\mathcal{U} = \operatorname{span}\{\boldsymbol{\delta}(\boldsymbol{w}) : \boldsymbol{w} \in \mathbb{F}^n\}, \qquad g = \dim \mathcal{U}.$$

For a fixed control, multiplying $\boldsymbol{\delta}(\boldsymbol{w})$ by a nonzero scalar merely rescales X and does not select a new direction. We therefore let n count only controls that remain independent after such rescaling is ignored; a control that only rescales the direction is not counted. Equivalently, the resulting family of direction lines is assumed to have n independent parameters. Such a family requires $g \geq n+1$, since the lines in a g-dimensional vector space have only $g-1$ independent parameters.

Now assume maximal partial-round rank growth and that every post-prefix direction lies in $\mathcal{K}_r(M)$. Then $\mathcal{U} \subseteq \mathcal{K}_r(M)$, while (4) gives $\dim \mathcal{K}_r(M) = t - r$. Hence

$$n+1 \leq g \leq t-r, \qquad \text{and therefore} \qquad r \leq t-n-1. \tag{34}$$

After the k input constraints have been built into the family, the remaining CICO conditions are the vanishing of the first k output coordinates:

$$F_{0,\boldsymbol{w}}(X) = \cdots = F_{k-1,\boldsymbol{w}}(X) = 0.$$

This is a system of k equations whose unknowns are the root variable X and the n controls in $\boldsymbol{w}$, hence $n+1$ unknowns in total. If these equations behave independently—in particular, if none is implied by the others and they have no unexpected common component—then each equation is expected to reduce the dimension of the solution set by one. Thus fewer than k unknowns gives a generically overdetermined system, while more than k unknowns is expected, when solutions exist, to leave free parameters. The dimension-balanced choice is therefore

$$n+1 = k, \qquad \text{equivalently} \qquad n = k-1.$$

This count is only a heuristic for the existence and dimension of the solution set. It neither guarantees a solution in the balanced case nor rules out an exceptional common zero in an overdetermined structured system. This equation count is the only heuristic step in the argument below.

Substituting $n+1 = k$ into (34) gives

$$k \leq g \leq t-r, \qquad \text{so} \qquad r \leq t-k.$$

If equality $r = t-k$ holds, then $\dim \mathcal{K}_{t-k}(M) = k$, and the inequalities force $g = k$. Because $\mathcal{U} \subseteq \mathcal{K}_{t-k}(M)$, equality of their dimensions then gives $\mathcal{U} = \mathcal{K}_{t-k}(M)$.

5.2 Matrix-Image Budget

The absorbed prefix is obtained by prescribing equations of the form $M\boldsymbol{v} = \boldsymbol{y}$. Each such equation fixes the image of one domain vector $\boldsymbol{v}$. If the prescribed domain vectors are jointly linearly independent, their images may be chosen independently; once their number reaches t, however, the action of M is fixed on a basis and no independent domain direction remains for completing the partial-round trail. The purpose of the following count is to determine how many of these basis directions are already consumed by the full-round prefix.

We count only the direct support-separated construction used for CICO-2 and CICO-3. It uses g nonempty, disjoint ratio groups, and the group components are chosen so that the resulting post-prefix direction space has dimension g. On a group, all dependence on the controls is carried by a scalar coefficient; the accompanying vector is fixed. A componentwise S-box changes that scalar coefficient but not the supporting vector. Consequently, the linear layer needs one prescribed image per fixed group vector, not one image per monomial in the controls.

The R_f matrix multiplications in the prefix contribute assignments in three parts. The first acts on

$$\boldsymbol{q}_0(x, \boldsymbol{w}) = \boldsymbol{d}_c + x\left(\boldsymbol{d}_x + \sum_{\ell=1}^{n} w_\ell \boldsymbol{d}_\ell\right).$$

It therefore requires the images of the constant vector $\boldsymbol{d}_c$ and the $n+1$ coefficient directions $\boldsymbol{d}_x, \boldsymbol{d}_1, \ldots, \boldsymbol{d}_n$, for $n+2$ assignments. After each of the next $R_f - 2$ S-boxes, the state is represented by one constant vector and g group-supported vectors, so the following matrix multiplication requires $g+1$ assignments. Before the last S-box of the prefix, the direct construction cancels the constant part. Its output therefore contains only the g group-supported vectors, and the last matrix multiplication requires g assignments. This is the same decomposition seen concretely as $3+3+3+2 = 11$ for CICO-2 and $4+4+4+3 = 15$ for CICO-3.

The group vectors are independent within any one stage because they are nonzero and have disjoint supports. Independence between vectors coming from different stages is not automatic. We therefore state it explicitly in the count below; it is a design condition verified by the two concrete constructions.

Proposition 4 (Direct prefix-assignment count). *Let $R_f \geq 2$. Consider the direct support-separated prefix described above, including the cancellation of its constant part before the last S-box, and assume that all domain vectors whose images it prescribes are jointly linearly independent. With n controls and g groups, the prefix then fixes M on*

$$N_{\mathrm{prefix}}(R_f, n, g) = (n+2) + (R_f - 2)(g+1) + g. \tag{35}$$

With the minimal post-prefix direction dimension $g = n+1$,

$$N_{\mathrm{prefix}}(R_f, n, n+1) = R_f(n+2) - 1. \tag{36}$$

If the subsequent trail-completion step needs at least one further domain vector whose image has not been fixed by the prefix, then necessarily

$$R_f(n+2) \leq t.$$

For the balanced choice $n = k-1$, this becomes

$$R_f(k+1) \leq t.$$

Proof. The three contributions counted above are $n+2$, $(R_f-2)(g+1)$, and g. Joint independence makes them additive, giving (35). Since $g \geq n+1$, the minimum occurs at $g = n+1$, where the sum is $R_f(n+2)-1$. Reserving one further independent domain vector requires $N_{\text{prefix}}+1 \leq t$, hence $R_f(n+2) \leq t$. Substituting $n = k-1$ gives $R_f(k+1) \leq t$.

This is a bookkeeping identity under the stated independence assumption, and the resulting inequality is only a necessary condition for this direct construction: the controls are carried through the full-round prefix by fixed vectors on disjoint ratio groups, the constant part is cancelled before the last prefix S-box, and all counted domain vectors are jointly independent. The inequality does not guarantee that the prescribed map can be completed to an invertible or MDS matrix with maximal rank growth and the desired characteristic-polynomial properties. Conversely, a different construction could deliberately introduce compatible dependencies among the prescribed domains and would not be governed by this count. The two explicit instances above verify the additional matrix properties for $k = 2$ and $k = 3$.

5.3 Degree Bounds at Width 16

Assume the maximal trail length $r_k = t-k$ is attained and the S-box exponent is α. After entry into the partial-round trail, the number of nonlinear layers that remain in the univariate representation is

$$L_k = R_f + R_P - \min(R_P, t-k) = R_f + \max(0, R_P - t + k).$$

Therefore

$$\deg F_{i,\boldsymbol{w}} \leq d_k := \alpha^{L_k}.$$

Equality additionally requires a nonzero leading coefficient.

For $t = 16$, $R_f = 4$, $R_P = 20$, and $\alpha = 3$,

$$r_k = 16 - k, \qquad L_k = 8 + k, \qquad d_k = 3^{8+k}.$$

Table 1 separates this degree bound from the independent matrix-image budget.

k	Controls	Groups	N_{prefix}	r_k	L_k	d_k	Matrix-Image Budget
1	0	1	7	15	9	3^9	fits
2	1	2	11	14	10	3^{10}	verified
3	2	3	15	13	11	3^{11}	verified
4	3	4	19	12	12	3^{12}	does not fit

Table 1. Direct four-full-round CICO-k construction for $t = 16$. One of the sixteen images is reserved for completion, so the prefix may prescribe at most fifteen.

For balanced CICO-4, the prefix alone would require $N_{\text{prefix}} = 4(4+1) - 1 = 19$ independent assignments, exceeding both the fifteen available with a

completion direction reserved and the sixteen available without one. Thus this row is not realizable by the direct, jointly independent construction; different relations or compatible dependencies among the prescribed vectors are not ruled out.

The hypothetical degree bound $d_4 = 3^{12} = 531441 < p = 2130706433$ concerns a separate issue. More generally, $3^{8+k} < p$ through $k = 11$, since $3^{19} < p < 3^{20}$. Having formal degree below the field size neither supplies the missing matrix assignments nor guarantees a common root; for CICO-4, the matrix-image budget already fails.

The inequality $d_k < p$ has a precise but limited meaning. Every function $\mathbb{F}_p \to \mathbb{F}_p$ has a unique representative of degree at most $p-1$ modulo $X^p - X$. If a nonzero formal polynomial already has degree below p, it has at most d_k base-field roots and its base-field root set is determined by

$$\gcd(F(X), X^p - X).$$

This does not guarantee that a root exists or that finding all roots is practical.

5.4 Finding CICO-k Solutions

The construction above produces a constrained-input family and lowers the degrees of its output equations; it does not by itself find a CICO-k solution. The remaining task is to select controls and solve the resulting univariate equations. We describe that step here because the degree bound determines the size of the polynomials passed to the root finder, but the solving methods themselves are prior art [4,3,15,17,18,2,16].

The root variable X is the only unknown left after a control $\boldsymbol{w} \in \mathbb{F}_p^{k-1}$ is fixed. One output equation can therefore be used to generate candidates, after which the remaining outputs serve as filters. Concretely, for each control:

1. construct the first constrained output polynomial $F_{0,\boldsymbol{w}}(X)$;
2. if this polynomial is nonzero, extract its base-field roots. A generic finite-field method first computes

$$H_{\boldsymbol{w}}(X) = \gcd(F_{0,\boldsymbol{w}}(X), X^p - X),$$

   whose linear factors are precisely the distinct roots of $F_{0,\boldsymbol{w}}$ in $\mathbb{F}_p$, and then splits $H_{\boldsymbol{w}}$;
3. for every root X_0, evaluate the complete permutation. Retain the candidate if $F_{i,\boldsymbol{w}}(X_0) = 0$ for $1 \le i < k$.

For NTT-friendly fields, the second step can instead use Graeffe-based root extraction [15,17,18]. Evaluating the complete permutation on the extracted roots verifies the other $k-1$ constraints without constructing their dense polynomial representations.

Let $\mathsf{C}_p(d)$ denote the cost of constructing a degree-at-most d first output polynomial, extracting its base-field roots, and testing the resulting candidates. This

cost depends on the root-extraction method as well as on d; for a Graeffe/NTT implementation it also depends on the smooth factors of $p-1$, available transform lengths, and memory-access costs [18]. If the degree bound is uniform over all control fibers and the root extractor is exhaustive, complete enumeration costs

$$O\big(p^{k-1}\mathsf{C}_p(d_k)\big)$$

field operations and covers every solution in the constructed p^k-point family.

Under the random-output heuristic, the first polynomial has approximately one base-field root on average, and a candidate satisfies the other $k-1$ constraints with probability $p^{-(k-1)}$. Across the p^{k-1} control values this predicts about one complete CICO-k solution, but it is an expectation rather than a complexity guarantee.

The accompanying SageMath implementation is available at

`https://github.com/s1fr0/slipway`

It searches fresh matrices and controls with configurable round counts and reference-generated constants. At $(p, t, \alpha, R_F, R_P) = (2130706433, 6, 3, 4, 5)$, it checks every matrix minor and solves the two output equations using a resultant in the post-prefix coefficients. The two initial full rounds are absorbed and four partial S-boxes are inactive, leaving degree 27. A separate verifier checks the supplied width-16 witness and its algebraic certificates; both modes use `poseidon-tools` to verify the complete permutation.

6 Security Considerations and Countermeasures

6.1 Matrix Checks

The 2026 Poseidon Cryptanalysis Initiative permits an arbitrary linear layer for the KoalaBear instance provided that it is MDS and satisfies the program's linked "no-invariant-subspace-trail" matrix check [6]. We ran these required checks on both M_2 and M_3. The MDS check was performed by verifying every nonempty square minor, as reported in the two concrete construction sections. Both matrices also return success under the matrix verifier linked by the Initiative.

The verifier implements the criteria of Grassi, Rechberger, and Schofnegger [10]. Its first branch uses their sufficient minimal-polynomial condition: it accepts when the minimal polynomial of M^i is irreducible of degree t for $1 \le i \le 2t$. Only if that sufficient test fails does the verifier run the three more detailed algorithms for invariant and iterative trails, including trails with active S-box coordinates and powers up to $4t$. For our matrices, the sufficient branch already applies.

The exact power checks establish

$$\chi_{M_k^i}(Z) \text{ is irreducible of degree } t \qquad \big(k \in \{2,3\},\ 1 \le i \le 4t\big). \tag{37}$$

An irreducible characteristic polynomial is also the minimal polynomial. If M_k^i had a nonzero proper invariant subspace, the restriction would give a nontrivial factor of its characteristic polynomial. Thus (37) excludes such invariant subspaces for every tested power of either concrete matrix. Moreover, because these characteristic polynomials are irreducible of full degree, they satisfy the sufficient minimal-polynomial branch of the Initiative's verifier. Equation (37) checks powers through $4t$, which is stronger than the $2t$ range required by that branch. This is why both M_2 and M_3 pass the linked matrix check without reaching the verifier's three fallback algorithms.

That property is compatible with the finite spaces $\mathcal{K}_r(M)$. Irreducibility excludes invariant subspaces, whereas $\mathcal{K}_r(M)$ describes a transient sequence of r hyperplane constraints. Every 16×16 matrix satisfies $\dim \mathcal{K}_{14}(M) \geq 2$. Maximal rank growth makes this space as small as possible but cannot remove it.

The MDS condition is likewise orthogonal: it controls the diffusion of sparse vectors through nonvanishing minors, whereas $\mathcal{K}_r(M)$ is defined by dense linear conditions on successive Krylov directions. A matrix can satisfy both properties, as the two concrete matrices demonstrate.

The missing property is full-prefix reachability. Matrix-only validation does not ask whether the fixed constants and a constrained-input family can satisfy the domain–image relations (12)–(17). Our construction exploits this joint relation between the constants and the subsequently selected matrix. It does not create an eighteen-round partial trail: its partial component has length fourteen, while the other four eliminated degree multipliers belong to the initial full rounds.

6.2 Scope

The result constructs a weak parameter instance; it is not an attack on a fixed standardized Poseidon matrix. The round constants are generated by the reference Grain LFSR, and the matrix is then constructed adaptively as a function of them. The construction therefore gives a class of matrices that satisfy the prescribed matrix-only checks but are weak relative to those constants. Consequently, the result applies to custom parameter generation that permits one component to be searched as a function of the other.

The concrete CICO-2 instance includes a complete input/output solution. The concrete CICO-3 instance establishes an MDS matrix, the three-dimensional trail-entry space $\mathcal{K}_{13}(M_3)$, and exact output degree, but it does not contain a common root. Likewise, the general CICO-k bounds have specific scopes. The bound $r \leq t - k$ assumes a balanced family whose k-dimensional post-prefix direction space lies in $\mathcal{K}_r(M)$, together with maximal partial-round rank growth. The bound $R_f(k+1) \leq t$ additionally assumes the direct prefix construction counted in Section 5.2: control dependence is carried on disjoint ratio groups, the prescribed domain vectors are jointly independent, and one further domain direction remains free for trail completion. Neither statement is a universal bound on other algebraic parametrizations or attacks.

6.3 Binding Constants to the Matrix

One way to prevent this adaptive dependence is to commit to M and derive the constants from a transcript containing a canonical encoding of M. For example,

$$T = \texttt{Poseidon-RC-v2} \,\|\, \mathrm{enc}(p, t, \alpha, R_F, R_P) \,\|\, \mathrm{enc}(M),$$
$$c_i^{(r)} = \mathrm{HashToField}_p\big(T \,\|\, \mathrm{enc}(r, i)\big).$$

Here enc is unique and length-delimited, matrix entries are encoded row by row in $[0, p-1]$, and $\mathrm{HashToField}_p$ uses domain-separated unbiased sampling from a cryptographic extendable-output function. Then equations such as $M\boldsymbol{\varepsilon}^{(2)} = -\boldsymbol{c}^{(3)}$ become hash-dependent fixed-point conditions rather than linear constraints that can be solved after the constants are known.

Equivalently, a specification may fix both a unique matrix and a unique constant sequence non-adaptively. Hash binding does not prevent grinding over complete transcripts; the transcript-selection rule must itself exclude adaptive search.

7 Conclusion

This paper identifies a class of Poseidon matrices that are weak relative to the round constants from which they are constructed. By choosing the matrix adaptively as a function of those constants, a constrained input family can be steered through the initial full rounds and directly into a finite partial-round trail. The support-separated construction prevents degree growth through the full-round prefix, an effect we call full-round absorption, while the trail prevents degree growth through the following partial rounds.

For the full KoalaBear instance of width 16, with a cubic S-box, eight full rounds, and twenty partial rounds, we find a complete CICO-2 solution: the first two coordinates of both the input and the output are zero.

The CICO-3 construction extends the method by adding a second control. Its matrix M_3 produces three variable directions for which the active S-box input remains constant through thirteen partial rounds. At the reported control, the first three output polynomials have degree 3^{11}.

We then generalize the construction to CICO-k and identify the limits imposed by the available trail directions and by the number of independent matrix actions required by the full-round prefix.

The result concerns adaptively generated parameter sets and is not an attack on a fixed standardized Poseidon instance. Both concrete matrices are MDS and pass the relevant linear-layer checks proposed by the Poseidon designers; those matrix-only checks do not test dependencies between a matrix and its round constants. Binding the constants to a canonical encoding of the matrix under a selection rule that excludes grinding, or fixing both components together non-adaptively, prevents the specific dependence used by the construction.

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Supplementary Material

A CICO-2 Matrix, Parameters, and Solution

Write $M_2 = (M_{2,L} \mid M_{2,R})$, where

$$M_{2,L} = \begin{pmatrix}
541074353 & 2067221306 & 435653776 & 643920239 & 1159448569 & 1705074076 & 523160259 & 2069510348 \\
124253918 & 1944393786 & 541476351 & 2081657340 & 1168784093 & 329353866 & 800491928 & 889329586 \\
1430081501 & 1231943135 & 335209979 & 1229299780 & 1969548122 & 2025282023 & 1096230993 & 787136366 \\
1258279241 & 1794765106 & 388624680 & 434968021 & 1748208215 & 2031798815 & 1884090105 & 1826024950 \\
802450403 & 96517910 & 1328895628 & 302770265 & 1607090936 & 2091296314 & 318209258 & 87304716 \\
1069997554 & 1683318092 & 729642910 & 1168297126 & 2000438927 & 1167842218 & 957988828 & 1619465014 \\
1730865959 & 1583666151 & 2119013864 & 1727267401 & 1232448640 & 1396502614 & 73659308 & 564894997 \\
599372201 & 263764922 & 1073195709 & 442175555 & 898623795 & 857789202 & 1173610968 & 274284313 \\
1324103846 & 1860436555 & 1985414053 & 1372253907 & 773371299 & 9569505 & 1943996097 & 1277083964 \\
1821684652 & 414881005 & 987742260 & 762977503 & 795778613 & 702439414 & 174971081 & 1594062874 \\
530473601 & 436073536 & 1243530224 & 293043868 & 335203772 & 182300910 & 1898928107 & 1626714363 \\
1150218008 & 1663311857 & 43573762 & 1471235180 & 450050282 & 1598258444 & 1207942683 & 1302838566 \\
859119476 & 2073553411 & 1643685798 & 1834423223 & 1823081958 & 907772738 & 510882258 & 844694866 \\
850428807 & 446781411 & 907146556 & 919919720 & 173309533 & 1130330074 & 873456441 & 487439945 \\
489609009 & 601171759 & 558315285 & 835764708 & 782299114 & 1089955541 & 1673102147 & 560842452 \\
321697498 & 740904349 & 1658944403 & 703974328 & 51339549 & 782054266 & 725034099 & 2002383317
\end{pmatrix}.$$

$$M_{2,R} = \begin{pmatrix}
712202895 & 1562752311 & 227852614 & 1705759745 & 1590626263 & 818373368 & 415683539 & 992321111 \\
855065037 & 1291068598 & 1382370648 & 532587292 & 1225193991 & 79282908 & 1083684740 & 803849014 \\
1045047588 & 1433196940 & 2083800676 & 1535053128 & 1906373109 & 838888220 & 2065671798 & 1813692563 \\
500290607 & 1424064667 & 322348595 & 1442656137 & 1235275688 & 1991909988 & 254989376 & 552047575 \\
792788649 & 831354109 & 805439566 & 592790001 & 815476797 & 1516642691 & 1181780200 & 1938752896 \\
939027853 & 869177199 & 2123665770 & 567764980 & 1664665021 & 1851884358 & 659416952 & 1143890128 \\
942033354 & 172485482 & 1430270513 & 1473629917 & 1011988127 & 1579359078 & 1683112462 & 1093583712 \\
435526980 & 729577898 & 37345520 & 1795356661 & 1280315324 & 1302107357 & 1072977794 & 506634067 \\
1720682607 & 1295539800 & 304321311 & 278778293 & 591347609 & 968766374 & 1893972479 & 1722951310 \\
455843891 & 1185934761 & 1920232802 & 1993788710 & 1699462130 & 1137208373 & 7892726 & 2089260364 \\
1466680270 & 1026338346 & 1856655024 & 2047312468 & 1747598382 & 1766381638 & 1300388862 & 1896345387 \\
103031339 & 1974679739 & 316585676 & 363520014 & 97465219 & 1197322690 & 1849930969 & 1407522799 \\
125457479 & 1164469748 & 395498273 & 1654855892 & 1280158498 & 1660719163 & 1249290307 & 345620430 \\
1618437958 & 346629786 & 430005257 & 1996446203 & 503227011 & 116970386 & 121168572 & 1529977258 \\
1600944998 & 1919437953 & 1922780853 & 820833148 & 1307676014 & 374260150 & 977816660 & 98254278 \\
1895915608 & 968467086 & 863710666 & 729961173 & 190353795 & 1862413256 & 1557969540 & 1182080540
\end{pmatrix}.$$

The input family (6) uses

$$\begin{aligned}\boldsymbol{d}_c = (&1646273927, 1846087681, 848418002, 1451245645,\\ &1874403116, 1526019671, 1044297521, 1150918050,\\ &1417510953, 955940910, 94674310, 1632384225,\\ &745706562, 211134455, 1502789393, 265897731)^T,\\ \boldsymbol{d}_x = (&0, 0, 1271726072, 1075483359, 1004956546, 1753720951,\\ &764237336, 152710851, 1644186190, 1112794565,\\ &530968735, 1624912103, 633463074, 2036570905,\\ &772263822, 422326334)^T,\\ \boldsymbol{d}_w = (&0, 0, 486203423, 866184324, 1517223536, 599947208,\\ &525119448, 147981569, 1491232208, 140285746,\\ &1024477059, 819765766, 844013187, 494237146,\\ &896642832, 2080909114)^T.\end{aligned}$$

The first two post-prefix directions are

$$\begin{aligned}\boldsymbol{u}_1 = (&0, 0, 0, 0, 0, 0, 0, 0, 0, 761240580, 673373400, 1563323388,\\ &987038617, 829652349, 1776810728, 2055687130)^T,\\ \boldsymbol{u}_2 = (&0, 1757960097, 1242623478, 1333779662, 1136155418,\\ &1958165781, 857968154, 95648885, 1628748799, 1294565886,\\ &1682337691, 1308451342, 1720941324, 1483226570,\\ &572726951, 599911387)^T.\end{aligned}$$

The six scalar ratios are

$$\begin{aligned}(\lambda_g^{(1)}, \lambda_h^{(1)}) &= (369934243, 1158169993),\\ (\lambda_g^{(2)}, \lambda_h^{(2)}) &= (1362970739, 1551057859),\\ (\lambda_g^{(3)}, \lambda_h^{(3)}) &= (715586279, 481984754).\end{aligned}$$

For the control and root reported in Section 3.4, the recovered input is

$$\begin{aligned}\boldsymbol{s}^\dagger = (&0, 0, 1994407524, 1518406169, 1869697938, 1680758970,\\ &1974616136, 2130375035, 5166025, 909317996, 1154992185,\\ &1870851706, 82420390, 1232840978, 1169612330, 1690942889)^T.\end{aligned}$$

Direct evaluation of the first four full rounds gives

$$\begin{aligned}\mathrm{Prefix}_4(\boldsymbol{s}^\dagger) = (&0, 939858986, 987921997, 1029076640,\\ &1362729673, 1749665979, 212186768, 1876370342, 902053652,\\ &671649736, 348972497, 1444167604, 372210245, 1435250072,\\ &631525605, 1256663773)^T,\end{aligned}$$

and the complete permutation gives

$$\begin{aligned}\pi(\boldsymbol{s}^{\dagger}) = (&0, 0, 1080431931, 1291539939, 842366831, 410517411,\\ &390312851, 1938087994, 1744025294, 601741681, 1822652989,\\ &1131652874, 158028812, 266015611, 365198253, 1770326166)^{T}.\end{aligned}$$

B CICO-3 Matrix and Parameters

In the concrete search, each ratio matrix $\Lambda^{(s)}$ was sampled with its first column normalized to 1, so the remaining entries in each row are ratios relative to the first direction. Singular matrices were rejected. The basis-change matrix P was sampled in the form specified in Section 4.1. A candidate was retained only when the prescribed domains were independent and the complete matrix construction passed the subsequent checks, including MDS.

Write $M_3 = (M_{3,L} \mid M_{3,R})$, where

$$M_{3,L} = \begin{pmatrix}
135881769 & 765759684 & 2009789902 & 67561320 & 164435303 & 836590230 & 1877251729 & 1923937313\\
626058966 & 614035633 & 18664249 & 1224773861 & 946087139 & 1608330272 & 1842633214 & 904655045\\
1779037118 & 164022131 & 1430748817 & 306186693 & 1168453629 & 283125483 & 1939987716 & 249426107\\
1802180895 & 157589721 & 419588439 & 1251560593 & 1649115318 & 1304935783 & 1952746135 & 175943624\\
1657254320 & 1494674635 & 1751018636 & 1365019446 & 2028897907 & 391289867 & 212649680 & 871080089\\
1581027245 & 1352172553 & 501328997 & 1572693656 & 1140320097 & 912392830 & 60489053 & 1932062183\\
1287973047 & 529852294 & 495669119 & 598014899 & 1855177438 & 1002028354 & 1729023125 & 1746544002\\
488788686 & 926486297 & 1228529182 & 1253951725 & 696583929 & 1452710825 & 635406545 & 734316253\\
1155042337 & 2022903479 & 2089297703 & 745797350 & 233629201 & 1611520275 & 1673370672 & 1389513436\\
1048837262 & 601686214 & 928064683 & 1609397228 & 89980036 & 1334970528 & 638921054 & 916626640\\
1074450454 & 1526556412 & 598549555 & 55573274 & 1748393782 & 1117075873 & 2110825697 & 961208115\\
671172782 & 184034401 & 1342889925 & 1935301208 & 348352611 & 208375733 & 2080689668 & 451080857\\
404257589 & 96596218 & 1623128002 & 1509940784 & 1028265528 & 1848042322 & 394156199 & 1867052741\\
1330566995 & 462233466 & 752942319 & 409183183 & 502283760 & 2059349370 & 1931144873 & 1046990846\\
1263953445 & 147476028 & 816844576 & 1084471280 & 1927100876 & 2033703553 & 1388152318 & 1282394446\\
2113546156 & 1586690560 & 1665348369 & 1047211900 & 552446858 & 908431534 & 1935053962 & 416089991
\end{pmatrix}.$$

$$M_{3,R} = \begin{pmatrix}
571746520 & 40266206 & 1979033653 & 309690800 & 1087853689 & 1769826427 & 1440515992 & 1871060926\\
1500951602 & 858982782 & 1390143614 & 130326710 & 1469988775 & 556434996 & 16635638 & 370225790\\
3508722 & 1816456387 & 1335055208 & 499440339 & 293356422 & 10759282 & 533984986 & 809464798\\
377097337 & 1103530178 & 2026484419 & 1777419778 & 2035772994 & 950866034 & 607201347 & 1347105738\\
978095341 & 2053590277 & 1073703090 & 1798840013 & 1773784979 & 2037325008 & 877158114 & 1904447226\\
1897995695 & 1081760016 & 1797634175 & 327861277 & 1401501911 & 1751963648 & 2106215261 & 1526428918\\
1349478493 & 722389716 & 154863522 & 761499357 & 2029471636 & 405273167 & 1562077698 & 546401538\\
737591430 & 1884381051 & 635602309 & 588810417 & 1801612898 & 1081201621 & 1685424560 & 210250590\\
874246977 & 687162623 & 1940648369 & 1261552433 & 976992813 & 2105104951 & 418367902 & 1646815907\\
1437689337 & 345547721 & 1105875334 & 1541802105 & 1473569243 & 308578489 & 608532826 & 461561443\\
81957444 & 1717203572 & 620301406 & 1126356920 & 1922226062 & 1345515398 & 1204985874 & 618786667\\
1189358280 & 482547847 & 730390406 & 781697900 & 902102119 & 1605853439 & 461316068 & 280517405\\
1989419670 & 1990091213 & 1293569636 & 759325974 & 1236251070 & 1113589096 & 1895919801 & 1989720393\\
1531209465 & 1549604589 & 1643064809 & 435801640 & 897751102 & 1577767837 & 10927054 & 1382802029\\
181332469 & 1641954542 & 1666809470 & 1184952357 & 941760065 & 1010655391 & 380499712 & 470839814\\
1424576654 & 530139879 & 617463401 & 955434045 & 671732672 & 1061238202 & 1908468122 & 92076336
\end{pmatrix}.$$

The input family (20) is defined by

$$\begin{aligned}
\boldsymbol{d}_c &= (1646273927, 1846087681, 1078133183, 987343865, 858781996, 889049418,\\
&\quad 108971947, 1100666405, 1112313699, 769765544, 1673511705, 1181761649,\\
&\quad 346344769, 503283102, 1204645159, 1309972698)^T,\\
\boldsymbol{d}_x &= (0, 0, 0, 815082177, 1580336821, 1391958076, 563717479, 287724130,\\
&\quad 1354153918, 657033816, 188095080, 1119757444, 1623165212, 775833032,\\
&\quad 943428807, 1090212786)^T,\\
\boldsymbol{d}_w &= (0, 0, 0, 38499303, 261152165, 1812283860, 1533193664, 1259230328,\\
&\quad 1012929408, 605291436, 1435283332, 1637197572, 262425658, 390920798,\\
&\quad 2005097720, 996006653)^T,\\
\boldsymbol{d}_z &= (0, 0, 0, 1572202524, 1011567234, 1122177016, 363150012, 1078271303,\\
&\quad 228449164, 1391483326, 1795329063, 432377298, 1806490718, 2075218277,\\
&\quad 1836826154, 1158066688)^T.
\end{aligned}$$

The three post-prefix directions are

$$\begin{aligned}
\boldsymbol{u}_1 &= (0, 0, 0, 1829900325, 1614448328, 168655764, 673377559, 268114019,\\
&\quad 839265115, 1461883355, 1295214259, 874964409, 896244049, 846303976,\\
&\quad 669507982, 281015670)^T,\\
\boldsymbol{u}_2 &= (0, 0, 0, 1256393336, 1938687327, 547117869, 1271330606, 1867245995,\\
&\quad 155980368, 1309672739, 657265951, 376277051, 1795601687, 994566849,\\
&\quad 1706504002, 400530890)^T,\\
\boldsymbol{u}_3 &= (0, 1469600577, 1713317349, 1469066977, 627915941, 1280173495,\\
&\quad 1170850004, 548782583, 763132694, 417214288, 1062857160, 391815185,\\
&\quad 2038031781, 2008534097, 532953884, 142296680)^T.
\end{aligned}$$

The ratio matrices used at the three support-separated stages are

$$\Lambda^{(1)} = \begin{pmatrix} 1 & 1563941991 & 1406597383 \\ 1 & 256438299 & 1412457815 \\ 1 & 1384217046 & 1382401229 \end{pmatrix}.$$

$$\Lambda^{(2)} = \begin{pmatrix} 1 & 940549499 & 1380419579 \\ 1 & 1358775160 & 45813488 \\ 1 & 193898430 & 1329799670 \end{pmatrix}.$$

$$\Lambda^{(3)} = \begin{pmatrix} 1 & 2078095724 & 999237501 \\ 1 & 1958642476 & 1462034771 \\ 1 & 1248924238 & 1885959254 \end{pmatrix}.$$

All three are invertible over $\mathbb{F}_p$. The final change from the sparse group basis to $\boldsymbol{b}_0, \boldsymbol{u}_1, \boldsymbol{u}_2$ uses

$$P = \begin{pmatrix} 438156930 & 306747013 & 1808727463 \\ 0 & 488913513 & 1704489083 \\ 0 & 2070389830 & 1638305142 \end{pmatrix}.$$

C Expanded CICO-3 Prefix Calculation

We expand the recurrence (25) under the assignments (26), using the notation of Section 4.1. For the first transition, prescribe

$$M_3\boldsymbol{d}_x = \boldsymbol{v}_1^{(1)}, \qquad M_3\boldsymbol{d}_w = \boldsymbol{v}_2^{(1)}, \qquad M_3\boldsymbol{d}_z = \boldsymbol{v}_3^{(1)}, \qquad M_3\boldsymbol{d}_c = \boldsymbol{e}^{(1)} - \boldsymbol{c}^{(1)},$$

where $\boldsymbol{e}^{(1)}$ is supported on $C^{(1)}$. After applying M_3 to $\boldsymbol{q}_0$ and adding the round-1 constants, the input to the next S-box is

$$\boldsymbol{\ell}_1 = \boldsymbol{e}^{(1)} + x \sum_{k=1}^{3} \gamma_k^{(0)} \boldsymbol{v}_k^{(1)}.$$

Applying (24) with $\boldsymbol{a} = \boldsymbol{\gamma}^{(0)}$ gives

$$\boldsymbol{\ell}_1^{\circ 3} = \boldsymbol{\varepsilon}^{(1)} + x^3 \sum_{g=1}^{3} \gamma_g^{(1)} \boldsymbol{z}_g^{(1)}. \tag{38}$$

The three group components produced by this first cubing now become the three incoming directions of the second transition. Prescribe

$$M_3\boldsymbol{z}_k^{(1)} = \boldsymbol{v}_k^{(2)} \quad (1 \le k \le 3), \qquad M_3\boldsymbol{\varepsilon}^{(1)} = \boldsymbol{e}^{(2)} - \boldsymbol{c}^{(2)},$$

where $\boldsymbol{e}^{(2)}$ is supported on $C^{(2)}$. After the matrix multiplication and addition of the round-2 constants,

$$\boldsymbol{\ell}_2 = \boldsymbol{e}^{(2)} + x^3 \sum_{k=1}^{3} \gamma_k^{(1)} \boldsymbol{v}_k^{(2)}.$$

A second application of (24) gives

$$\boldsymbol{\ell}_2^{\circ 3} = \boldsymbol{\varepsilon}^{(2)} + x^9 \sum_{g=1}^{3} \gamma_g^{(2)} \boldsymbol{z}_g^{(2)}. \tag{39}$$

For the final S-box of the prefix, we remove the constant part completely and prescribe

$$M_3\boldsymbol{z}_k^{(2)} = \boldsymbol{v}_k^{(3)} \quad (1 \le k \le 3), \qquad M_3\boldsymbol{\varepsilon}^{(2)} = -\boldsymbol{c}^{(3)}.$$

The second equality cancels the round-3 constants, so the S-box input is

$$\boldsymbol{\ell}_3 = x^9 \sum_{k=1}^{3} \gamma_k^{(2)} \boldsymbol{v}_k^{(3)}.$$

Cubing once more gives

$$\boldsymbol{\ell}_3^{\circ 3} = x^{27} \sum_{g=1}^{3} \gamma_g^{(3)} \boldsymbol{z}_g^{(3)}. \tag{40}$$

At this point the three full S-boxes have affected only the scalar column

$$\boldsymbol{\gamma}^{(0)} \longmapsto \boldsymbol{\gamma}^{(1)} \longmapsto \boldsymbol{\gamma}^{(2)} \longmapsto \boldsymbol{\gamma}^{(3)};$$

the vector part has remained a combination of exactly three directions.

D MDS Lower Bound on Output-Group Size

The following lemma generalizes the support argument in Section 4.2 to any number of groups.

Lemma 3 (MDS lower bound on output-group size). *Let $M \in \mathbb{F}^{t\times t}$, and let $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_m \in \mathbb{F}^t$ be supported together on a coordinate set S of size at most σ; that is,*

$$\bigcup_{j=1}^{m} \operatorname{supp}(\boldsymbol{s}_j) \subseteq S, \qquad |S| \le \sigma.$$

Partition the output coordinates into nonempty groups $G_1, \ldots, G_m$. For each G_ℓ, let $\boldsymbol{r}_\ell \neq 0$ be supported on G_ℓ. Suppose the images of the inputs have the form

$$M\boldsymbol{s}_j = \sum_{\ell=1}^{m} \lambda_{\ell,j} \boldsymbol{r}_\ell \qquad (1 \le j \le m), \tag{41}$$

where the ratio matrix

$$\Lambda = \left(\lambda_{\ell,j}\right)_{\ell,j=1}^{m} \in \mathrm{GL}_m(\mathbb{F}).$$

Thus, on group G_ℓ, all the images have the same coordinate pattern $\boldsymbol{r}_\ell$ and differ only by the ratios in row ℓ of Λ; column j describes the image of $\boldsymbol{s}_j$.

If M is MDS, then every output group satisfies

$$|G_\ell| \ge t + 1 - \sigma \qquad (1 \le \ell \le m).$$

Proof. Fix G_ℓ. Since Λ is invertible, choose

$$\boldsymbol{\mu} = \Lambda^{-1} e_\ell$$

where here e_ℓ is the ℓ-th standard basis vector of $\mathbb{F}^m$, and form the input combination

$$\boldsymbol{x} = \sum_{j=1}^{m} \mu_j \boldsymbol{s}_j.$$

Every $\boldsymbol{s}_j$ is supported on S, so $\mathrm{supp}(\boldsymbol{x}) \subseteq S$ and therefore $\mathrm{wt}(\boldsymbol{x}) \leq \sigma$.

Using (41),

$$\begin{aligned} M\boldsymbol{x} &= \sum_{j=1}^{m} \mu_j M \boldsymbol{s}_j \\ &= \sum_{a=1}^{m} (\Lambda\boldsymbol{\mu})_a \boldsymbol{r}_a \\ &= \boldsymbol{r}_\ell, \end{aligned}$$

because $\Lambda\boldsymbol{\mu} = e_\ell$. Since $\boldsymbol{r}_\ell \neq 0$, this also shows that $\boldsymbol{x} \neq 0$. Moreover, $\mathrm{wt}(M\boldsymbol{x}) = \mathrm{wt}(\boldsymbol{r}_\ell) \leq |G_\ell|$. The MDS branch-number inequality now gives

$$t + 1 \leq \mathrm{wt}(\boldsymbol{x}) + \mathrm{wt}(M\boldsymbol{x}) \leq \sigma + |G_\ell|,$$

which rearranges to the claimed lower bound.